# Enhanced Total Variation Regularized Representation Model With Endmember Background Dictionary for Hyperspectral Anomaly Detection

Chunhui Zhao, Chuang Li, *Student Member, IEEE*, Shou Feng, *Member, IEEE*, and Xiuping Jia, *Fellow, IEEE*

***Abstract*— In recent years, several representation models based on total variation (TV) have been proposed for hyperspectral imagery (HSI) anomaly detection. However, the TV terms of these works are directly imposed on the representation coefficient matrix, which can destroy the spatial structure of an HSI to some extent. Besides, as the spatial resolution of an HSI is relatively low, mixed pixels existing in an HSI can lead to anomaly component contamination, which can make the difference between background and anomalies not significant enough. To address these issues, a novel enhanced TV (ETV) with an endmember background dictionary (EBD) for hyperspectral anomaly detection is proposed. The ETV is designed to be used on the row vectors of the representation coefficient matrix to enhance the spatial structure of an HSI in the presentation process. Furthermore, the proposed ETV regularized representation model with EBD (ETVEBD) method elaborates on a background dictionary constructed by endmembers of background pixels, which are pure spectral signatures of background pixels. The proposed EBD can decrease the influence of anomaly components in mixed pixels, and the coefficient matrix of the EBD has more physical meanings. The proposed method is evaluated on four hyperspectral datasets, and the experiment results show that its performance is the best compared with the other seven state-of-the-art methods.**

***Index Terms*— Endmember background dictionary (EBD), enhanced total variation (ETV), hyperspectral anomaly detection, representation model.**

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## I. Introduction

OVER past few decades, hyperspectral remote sensing has drawn lots of attention for recognizing ground surface materials [1]. As the 3-D data cube with hundreds or even thousands of spectral bands, the hyperspectral imagery (HSI) can provide the spectral and spatial characteristics of different objects of interest [2]. Therefore, there are many branches of the hyperspectral remote sensing research, such as anomaly detection [3], target detection [4], and classification [5].

Hyperspectral anomaly detection actually classifies every pixel under test (PUT) as background or anomaly, which is also regarded as an unsupervised binary classification problem. Anomalous targets are usually defined as rare and special objects whose spectra are completely different from the surrounding backgrounds. Due to the nature of requiring no prior information, namely, no need for spectrums or labels of backgrounds and anomalies, anomaly detection has been implemented in various application fields, such as military surveillance [6], search-and-rescue [7], mineral mapping [8], and food quality and safety control [9].

Due to the efforts of previous researchers, lots of anomaly detection methods have been proposed. Among these methods, the Reed–Xiaoli (RX) detector, known as the benchmark of hyperspectral anomaly detection, is the most popular method, which is built on the statistical model [10]. It assumes that the backgrounds follow the multivariate normal distribution and make use of the generalized likelihood ratio test decision rule, which is simplified as the Mahalanobis distance of the PUT and backgrounds. The drawback of the RX detector is that the anomalies and noises embedded in backgrounds can decrease its detection accuracy. In order to overcome this problem, some improved versions of the RX detector have been proposed. Both blocked adaptive computationally efficient outlier nominator (BACON) [11] and the random-selection-based anomaly detector (RSAD) [12] adopted the iteration strategy to mitigate the contamination of the backgrounds. Besides, the cluster-based anomaly detector (CBAD) [13] and the segmented-RX [14] divide the HSI into some clusters to obtain the pure homogeneous regions. However, the methods aforementioned are all based on the assumption that the

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backgrounds obey the multivariate normal distribution. However, due to the complexity of different land covers, the multivariate normal distribution assumption is not suitable to model the real-world HSIs, which always include numerous classes of materials. Therefore, the kernel RX (KRX) detector [15] based on the kernel theory adequately models the backgrounds by projecting the HSI into the high-dimensional space. Moreover, by using robust regression analysis based on kernel theory, a robust nonlinear anomaly detection (RNAD) method was proposed in [16].

Different from the statistic model, the representation-based approaches have been well-developed recently, which do not need the multivariate normal distribution assumption. The philosophy of these algorithms is that the background data can be approximately represented by the constructed background dictionary, while the anomalies cannot. The collaborative-representation-based detector (CRD) is a typical representative of this type of approach [17]. It obtains the spatial neighborhood pixels of the PUT as the background dictionary and takes advantage of the $l_2$-norm minimization to reinforce the collaboration of representation vectors. Similar to the KRX detector, the CRD detector further is extended into a kernel version, i.e., the kernel CRD (KCRD). Moreover, with the development of the sparse representation theory, sparse representation-based detectors have also been proposed for hyperspectral anomaly detection. Specifically, the background joint sparse representation (BJSR) detector adaptively selects the background bases for local regions [18]. From the perspective of the frequency of the dictionary atoms used for HSI reconstruction, a sparsity score estimation framework for hyperspectral anomaly detection was proposed [19].

Nowadays, the low-rank-based methods for hyperspectral anomaly detection have achieved great success. As a subspace recovery technique, the robust principal component analysis (RPCA) [20] decomposes an HSI into low-rank and sparse components [21]. Based on RPCA, RPCA-RX [22], respectively, models the backgrounds by low-rank constraint and anomalies by sparse constraint and applies the RX detector on the sparse component to detect anomalies. Nonetheless, the RPCA does not take into account the influence of the noise, which may increase the false alarm rate (FAR). Thus, to reduce the impact of noise, the low-rank and sparse matrix decomposition (LRaSMD) separates the noise from the low-rank component and sparse component [23]. Then, the LRaSMD-based Mahalanobis distance (LSMAD) method computes the Mahalanobis distance between the low-rank component decomposed by LRaSMD and the PUT as the detection value [24]. Since there are numerous mixed pixels in an HSI, the assumption that the HSI data lies in a single subspace may be inaccurate. Therefore, the low-rank representation (LRR) [25] exploits a dictionary to construct the multiple subspaces that an HSI is assumed to lie in. Meanwhile, the low-rank and sparse representation (LRASR)-based detector applied the LRR to hyperspectral anomaly detection for the first time [26]. To improve the performance of the LRR-based detector, some methods also have been proposed in some closer years. For example, the abundance- and dictionary-based low-rank (ADLR) decomposition detector applied spectral unmixing to obtain the abundance vectors and utilized these abundance vectors to detect anomalies [27]. It is worth mentioning that the influence of the noise will be reduced by using abundance vectors to detect anomalies in the ADLR method.

What is more, in order to take advantage of the spatial information in an HSI, some methods that incorporate both spatial information and spectral information have also been proposed. As a powerful regularizer, the total variation (TV) term that can preserve the edge of different materials and promote piecewise smoothness has been successfully applied in some hyperspectral image processing fields, such as unmixing [28], denoising [29], and clustering [30]. Considering the spatial relationships of each PUT and its corresponding neighbors, Cheng and Wang [31] have employed the TV regularizer to the representation model for hyperspectral anomaly detection. Similarly, Feng *et al.* [32] proposed a novel local spatial constraint and TV detector for hyperspectral anomaly detection. Besides, the prior-based tensor approximation (PTA) method [33] decomposes the HSI into a background tensor and an anomaly tensor, and the piecewise smooth prior is embedded by the linear TV regularizer.

However, since the original HSI and the background dictionary are represented as 2-D matrices, the spatial structure of hyperspectral data will be lost to some extent when the TV is directly used to regularize the representation coefficient matrix. Furthermore, due to the low spatial resolution of HSI, the existence of mixed pixels is also an important factor that cannot be ignored. The mixed pixels that exist in the background dictionary can affect the reconstruction errors, which are used to detect anomalies. That is to say, when the background dictionary contains the mixed pixels, the reconstruction errors of the pixels that are difficult to detect are not large enough. In addition, the coefficients of previous representation-based detectors cannot show the abundances of mixed pixels [34].

To address the above issues, we propose a novel enhanced TV (ETV) regularized representation model with an endmember background dictionary (ETVEBD) for hyperspectral anomaly detection in this article. To avoid the loss of the spatial structure of the HSI, a novel ETV term and the $l_{2,1}$ sparsity regularizer are used to facilitate the piecewise smoothness of background components and sparsity of the anomaly components, respectively. The ETVEBD method first reshapes each row of the representation coefficient matrix into a matrix, which is called the enhanced representation coefficient (ERC) matrix, and imposes the TV regularizer on the ERC matrix. Furthermore, to get rid of the anomaly components in mixed pixels of the background dictionary, an endmember background dictionary (EBD) construction method is proposed. It is noteworthy that the ETVEBD algorithm integrates the ETV regularized representation model and the EBD into a unified framework. Hence, the representation coefficients in the ETVEBD algorithm have more physical meanings.

The main contributions of the proposed ETVEBD method can be summarized as follows.

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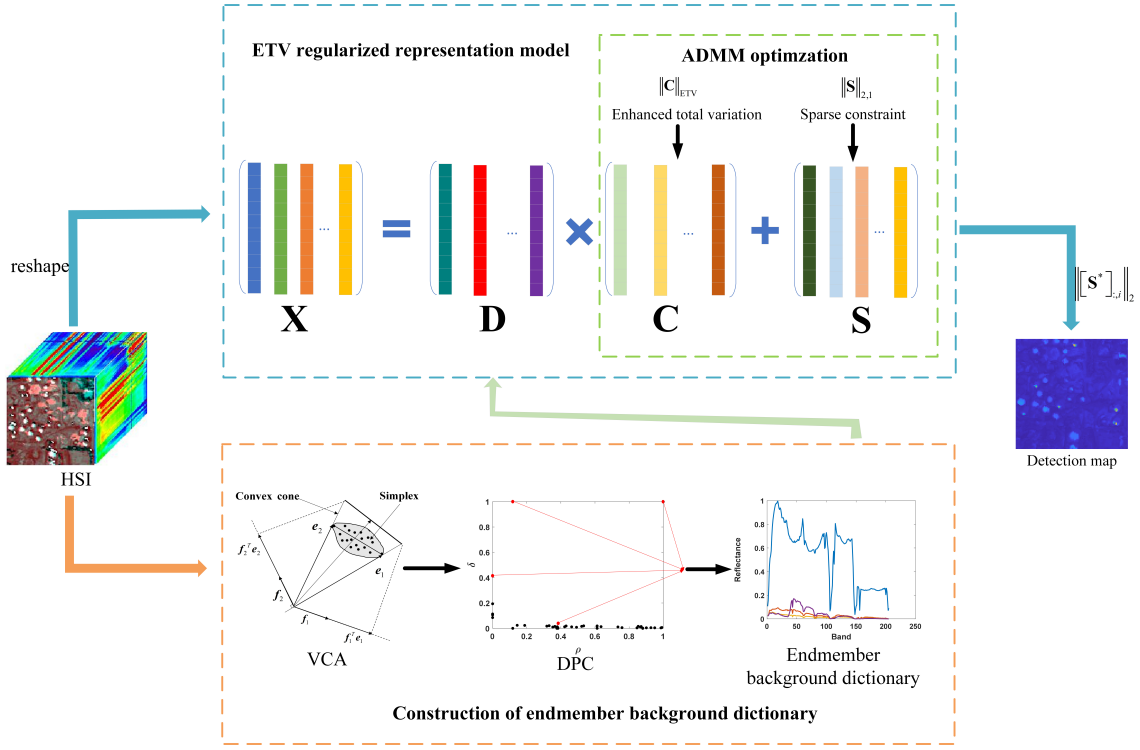


Fig. 1. Framework of the proposed method.

1) A novel ETV term is proposed to make the dictionary atoms efficiently represent backgrounds. Then, the ETV term is integrated into the model to represent the background with the enhanced spatial structure of the HSI, and the $l_{2,1}$-norm sparsity regularizer is used to characterize the anomalies. In this way, compared with the traditional TV term regularized model, the spatial structure of the HSI can be enhanced in ETVEBD.
2) To reduce the anomaly components contamination of the background dictionary, an EBD construction method is proposed. In consideration of the computation complexity, a simple and fast algorithm, vertex component analysis (VCA) [35], is used to extract the endmembers from the original HSI to construct a background dictionary. Then, the so-called density peak-based clustering (DPC) [36] method, which does not need any parameter, is employed to eliminate the anomaly endmembers. The proposed EBD construction method can not only reduce the anomaly components of mixed pixels in the background dictionary but also can preserve the physical meanings of the representation coefficients.

The rest of this article is organized as follows. Section II introduces the proposed ETVEBD method in detail. Section III presents the experimental datasets and the experimental results. The conclusion is drawn in Section IV.

## II. Proposed ETVEBD Method

The framework of the proposed ETVEBD method is given in Fig. 1. As shown in Fig. 1, the background endmembers are first extracted to form the EBD. Then, the ETVEBD method

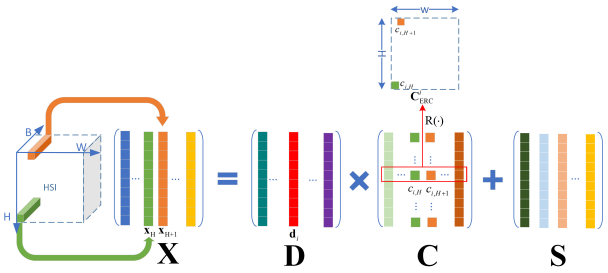


Fig. 2. Illustration of the spatial structure lost in the traditional TV regularized representation model and the enhanced spatial structure in the ETVEBD method.

is optimized by the alternating direction method of multipliers (ADMM) [37], and the minimizer $\mathbf{S}^*$ will be obtained to detect anomalies. In the following, each step of the ETVEBD method will be introduced.

### A. Enhanced Total Variation Regularized Representation Anomaly Detection Model

Fig. 2 intuitively illustrates the spatial structure lost in the traditional TV regularized representation model. To be specific, for an element $c_{i,j}$ of the representation coefficient matrix $\mathbf{C}$, it represents the representation coefficient of the dictionary atom $\mathbf{d}_i$ corresponding to the PUT $\mathbf{x}_j$. When a 3-D HSI is reshaped to the 2-D matrix, some nonadjacent pixels will be rearranged into adjacent ones, such as the pixel $\mathbf{x}_{\mathrm{H}}$ and $\mathbf{x}_{\mathrm{H}+1}$. Then, for the same dictionary atom $\mathbf{d}_i$, its representation coefficients $c_{i,\mathrm{H}}$ and $c_{i,\mathrm{H}+1}$ corresponding to the pixels $\mathbf{x}_{\mathrm{H}}$ and $\mathbf{x}_{\mathrm{H}+1}$ will be promoted to be similar. In fact, since the pixels

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$\mathbf{x}_{\mathrm{H}}$ and $\mathbf{x}_{\mathrm{H}+1}$ are not adjacent, the representation coefficients $c_{i,\mathrm{H}}$ and $c_{i,\mathrm{H}+1}$ should not be similar.

As described above, if the TV regularizer is directly applied to the representation coefficient matrix, the spatial structure of an HSI will be destroyed. In view of this, Zhai *et al.* [30] proposed a novel TV regularized collaborative representation clustering method with a locally adaptive dictionary (TV-CRC-LAD) for hyperspectral clustering. Concretely, similar to the spectral unmixing research, each row of representation coefficient matrix can be seen as a specific "fractional abundance." If each row of the representation coefficient matrix is reshaped into a 2-D matrix, it can be seen as a fractional abundance map. Then, the TV regularizer is employed on the "fractional abundance" matrix to promote piecewise smoothness. Inspired by the TV-CRC-LAD method, the ETV term regularized representation model for hyperspectral anomaly detection is proposed to enhance the spatial information of the proposed representation model for hyperspectral anomaly detection.

Let $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^{H\times W} \in \mathbb{R}^{B\times H\times W}$ be a 3-D HSI data cube, where $B$ is the number of spectral bands, $H$ is the height of the data cube, $W$ is the width of the data cube, and $\mathbf{x}_i$ is the $i$th spectral vector that can be written as $\mathbf{x}_i = (x_i^1, x_i^2, \ldots, x_i^B)^T$. Denote the total number of pixels in $\mathbf{X}$ as $N$, where $N = H\times W$, and $\mathbf{X}$ can be rewritten as a 2-D matrix whose each row represents a band image and a column represents a pixel. The optimization problem of the proposed model can be formulated as follows:

$$\min_{\mathbf{C},\mathbf{S}} \ \frac{1}{2}\|\mathbf{X}-\mathbf{D}\mathbf{C}-\mathbf{S}\|_F^2 + \beta\|\mathbf{C}\|_{\mathrm{ETV}} + \lambda\|\mathbf{S}\|_{2,1} \tag{1}$$

where $\mathbf{D} = [\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_n]$ is the endmember dictionary matrix, with each column representing a dictionary atom. $\mathbf{C} = [\mathbf{c}_1, \mathbf{c}_2, \ldots, \mathbf{c}_N]$ is the representation coefficient matrix, and $\mathbf{S}$ denotes the remaining part, i.e., the anomaly part, respectively. $\beta$ and $\lambda$ are two regularization parameters that balance the contribution of each term, $\|\cdot\|_{2,1}$ is the $l_{2,1}$-norm that can model the sparsity of sample-specific anomalies, $\|\cdot\|_F$ is the Frobenius-norm that is used to define the data fidelity term, i.e., $(1/2)\|\mathbf{X}-\mathbf{D}\mathbf{C}-\mathbf{S}\|_F^2$, $\|\cdot\|_{\mathrm{ETV}}$ is the proposed ETV term. It is worth noting that how to construct the endmember dictionary $\mathbf{D}$ is introduced in Section II-B. The definition of the TV term is given as follows:

$$\|\mathbf{C}\|_{\mathrm{TV}} = \sum_{\{i,j\}\in\varepsilon} \|\mathbf{c}_i - \mathbf{c}_j\|_1 \tag{2}$$

where $\mathbf{c}_i$ and $\mathbf{c}_j$ represent the $i$th and $j$th coefficient vectors of representation coefficient matrix $\mathbf{C}$, respectively, and the $\varepsilon$ denotes the first-order adjacent vectors (in the horizontal and vertical directions), which means that there are two adjacent vectors for each $\mathbf{c}_i$, $\|\cdot\|_{\mathrm{TV}}$ denotes the TV norm regularizer [38].

Denote a row vector of $\mathbf{C}$ as $\mathbf{C}_{i,:}$; the ETV term reshapes $\mathbf{C}_{i,:}$ into the ERC matrix $\mathbf{C}_i^{\mathrm{ERC}}$ with the size of $H\times W$. Then, the proposed ETV term can be defined as follows:

$$\|\mathbf{C}\|_{\mathrm{ETV}} = \sum_{i=1}^{n} \left\|\mathbf{C}_i^{\mathrm{ERC}}\right\|_{\mathrm{TV}} = \sum_{i=1}^{n} \|\mathrm{R}(\mathbf{C}_{i,:})\|_{\mathrm{TV}} \tag{3}$$

where $\mathrm{R}(\cdot)$ is the reshaping operator, which reshapes the row vector of the representation coefficient matrix to the ERC matrix. As shown in Fig. 2, taking advantage of the reshaping operator $\mathrm{R}(\cdot)$, the representation coefficients $c_{i,\mathrm{H}}$ and $c_{i,\mathrm{H}+1}$ are restored to the same position as pixels $\mathbf{x}_{\mathrm{H}}$ and $\mathbf{x}_{\mathrm{H}+1}$ in original 3-D HSI, respectively. The row vector of the representation coefficient matrix in which $c_{i,\mathrm{H}}$ and $c_{i,\mathrm{H}+1}$ are located in will be reshaped to the ERC matrix. The physical meanings of $\mathbf{C}_i^{\mathrm{ERC}}$ are that the element $c_{i,j}^{\mathrm{ERC}}$ of the ERC matrix $\mathbf{C}_i^{\mathrm{ERC}}$ represents the representation coefficient of the dictionary atom $\mathbf{d}_i$ corresponding to the PUT $\mathbf{x}_{i,j}$, which is located at the position $(i, j)$ of the original 3-D HSI. Thus, if a TV regularizer is imposed on the ERC matrix to promote its piecewise smoothness, the spatial structure of HSI will be enhanced compared with the original TV regularized representation model.

### *B. Construction of Endmember Background Dictionary*

Almost all the representation models for hyperspectral anomaly detection utilize the background dictionary constructed by background pixels without considering the mixed pixels [39]–[41]. In fact, because of the low spatial resolution of hyperspectral sensors, there are a lot of mixed pixels in an HSI. The existence of mixed pixels in the background dictionary can make it polluted by anomaly components in mixed pixels, and reconstruction errors of pixels will be not large enough to distinguish anomalies from backgrounds. Furthermore, the representation coefficients do not have any physical meaning. To this end, a new EBD construction method is proposed in this part.

In the hyperspectral unmixing field, the definition of endmembers is the pure or relatively pure spectra of materials [42]. In consideration of the computation complexity, the VCA method is used to extract the endmembers. The VCA algorithm is based on the geometry of convex sets and assumes that the endmembers occupy the vertices of a simplex. By iteratively projecting data onto a direction orthogonal to the subspace spanned by the endmembers already extracted, a new endmember signature corresponding to the extreme of the projection will be found. Furthermore, to make the ETVEBD method more practical, we first estimated the class number of the ground materials in HSI, i.e., $c$ by the well-known Hysime algorithm [43]. Besides, to ensure that ETVEBD can extract all kinds of endmembers, we set $k = 1.5c$ in the experiments, where $k$ is the number of endmembers extracted by the VCA method. Let $\mathbf{E} = \{\mathbf{e}_i\}_{i=1}^{k} \in \mathbb{R}^{B\times k}$ denote the endmember matrix extracted by the VCA algorithm. Note that the extracted endmember matrix $\mathbf{E}$ may have the endmembers belong to anomaly targets. Then, the DPC method is employed to eliminate the anomaly endmembers.

Unlike the $k$-means [26], mean shift [44] or density-based spatial clustering of applications with noise (DBSCAN) algorithm [45], the DPC algorithm does not need presetting parameters. The DPC method is only based on the similarity between the data points. It assumes that the cluster centers are surrounded by neighbor data points with lower local density, and the cluster centers also have a relatively large distance from other samples with higher density. Given a dataset $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^{N} \in \mathbb{R}^{B\times N}$ with $N$ samples of $B$ dimensions, the

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local density $\rho_i$ can be calculated as follows:

$$\rho_i = \sum_j \chi(d_{i,j} - d_c) \tag{4}$$

where $d_{i,j}$ is the Euclidean distance between the point $\mathbf{x}_i$ and $\mathbf{x}_j$, and $d_c$ is the cutoff distance whose calculation formulation is

$$d_c = \mathbf{d}_s(p), \quad p = \langle 0.02 \times k \times (k-1) \rangle \tag{5}$$

where $\mathbf{d}_s$ is ascending sorted result of $d_{i,j}$, $\langle \cdot \rangle$ is the round operation, and $\chi(\cdot)$ is the indicator function, which can be defined as follows:

$$\chi(x) = \begin{cases} 1, & \text{if } x < 0 \\ 0, & \text{if } x \geq 0. \end{cases} \tag{6}$$

Then, $\delta_i$ means that the minimum distance between data point $x_i$ and the points with larger local density. It can be defined as follows:

$$\delta_i = \min_{j:\rho_j > \rho_i} (d_{i,j}). \tag{7}$$

According to the assumption of the DPC method, the data points with large values of $\delta$ and relatively large values of $\rho$ can be chosen as the cluster centers. Hence, the cluster centers can be identified by the scores defined as

$$\gamma_i = \rho_i \times \delta_i. \tag{8}$$

In addition, cluster centers can be selected as the atom to form the background dictionary [46]. That is to say, the pixel with a larger value of $\gamma$ has a greater probability to be a background pixel, and the endmember background dictionary can be built according to the value of $\gamma$ of each endmember. In particular, for each endmember $\mathbf{e}_i$ obtained by the VCA algorithm, the ETVEBD calculates its corresponding $\gamma_i$. Then, we sort them in descending order, and top-$n$ corresponding endmembers are chosen to form the endmember dictionary. $n$ is calculated by the following formula:

$$n = \langle p \times k \rangle \tag{9}$$

where $p$ is the percentage of the endmembers selected to form the endmember dictionary, and $\langle \cdot \rangle$ is the round operation. The procedure of EBD construction is summarized in Algorithm 1.

**Algorithm 1** EBD Construction Algorithm

**Input:**

$\mathbf{X}$: 2-D hyperspectral data; $p$: the percentage of the endmembers selected to form the endmember background dictionary.

**Output:**

Endmember background dictionary $\mathbf{D}$.

**Initialize:**

$\mathbf{D} = \varnothing$

Extract the endmembers $\mathbf{E} = \{\mathbf{e}\}_{i=1}^{k} \in \mathbb{R}^{B \times k}$.
**for** $i = 1 : k$ **do**
  Computing the $\gamma_i$ via (8).
**end for**
Sort $\gamma_i$ in desecending order.
Compute the $n$ via (9).
Add the pixels with top $n$, $\gamma$ values into the dictionary $\mathbf{D}$.

### C. Optimization Procedure

By introducing an auxiliary variable $\mathbf{V}$, the optimization problem (1) can be converted into the following equivalent problem:

$$\min_{\mathbf{C},\mathbf{S},\mathbf{V}} \frac{1}{2}\|\mathbf{X} - \mathbf{DC} - \mathbf{S}\|_F^2 + \beta\|\mathbf{V}\|_{\text{ETV}} + \lambda\|\mathbf{S}\|_{2,1} \quad \text{s.t. } \mathbf{V} = \mathbf{C}. \tag{10}$$

The augmented Lagrangian function of the optimization problem (10) can be formulated as follows:

$$L(\mathbf{C},\mathbf{S},\mathbf{V},\mathbf{M},\mu) = \frac{1}{2}\|\mathbf{X} - \mathbf{DC} - \mathbf{S}\|_F^2 + \beta\|\mathbf{V}\|_{\text{ETV}} + \lambda\|\mathbf{S}\|_{2,1} + \frac{\mu}{2}\|\mathbf{V} - \mathbf{C} - \mathbf{M}\|_F^2 \tag{11}$$

where $\mu$ is the penalty parameter and $\mathbf{M}$ is the Lagrange multiplier. Then, the minimization of (11) can be solved by ADMM. Specifically, (11) can be split into three subproblems as follows.

*1) Updating of* $\mathbf{C}$*:* The subproblem can be formulated as follows:

$$\mathbf{C}^{t+1} = \arg\min_{\mathbf{C}} \frac{1}{2}\|\mathbf{X} - \mathbf{DC} - \mathbf{S}^t\|_F^2 + \frac{\mu^t}{2}\|\mathbf{V}^t - \mathbf{C} - \mathbf{M}^t\|_F^2 \tag{12}$$

where $t$ is the index of iteration and the subproblem (12) has a closed-form solution, as shown in the following:

$$\mathbf{C}^{t+1} = (\mathbf{D}^T\mathbf{D} + \mu^t\mathbf{I})^{-1}[\mathbf{D}^T(\mathbf{X} - \mathbf{S}^t) + \mu^t(\mathbf{V}^t - \mathbf{M}^t)]. \tag{13}$$

*2) Updating of* $\mathbf{S}$*:* The subproblem of $\mathbf{S}$ can be formulated as follows:

$$\mathbf{S}^{t+1} = \arg\min_{\mathbf{S}} \frac{1}{2}\|\mathbf{X} - \mathbf{DC}^{t+1} - \mathbf{S}\|_F^2 + \lambda\|\mathbf{S}\|_{2,1} \tag{14}$$

which has a closed-form solution by using the $l_{2,1}$-norm minimization operator [25]

$$\mathbf{S}^{t+1} = \mathbf{\Omega}_{2,1}(\mathbf{X} - \mathbf{DC}^{t+1}, \lambda) \tag{15}$$

where $\mathbf{\Omega}$ is the $\ell_{2,1}$-minimization operator.

*3) Updating of* $\mathbf{V}$*:* The subproblem of $\mathbf{V}$ can be formulated as follows:

$$\mathbf{V}^{t+1} = \arg\min_{\mathbf{V}} \beta\|\mathbf{V}\|_{\text{ETV}} + \frac{\mu^t}{2}\|\mathbf{V} - \mathbf{C}^{t+1} - \mathbf{M}^{t+1}\|_F^2. \tag{16}$$

For the optimization problem (16), the fast gradient-based algorithm can solve it efficiently [47].

*4) Updating of* $\mathbf{M}$*:* During the optimization process, the Lagrangian multiplier $\mathbf{M}$ can be updated by as follows:

$$\mathbf{M}^{t+1} = \mathbf{M}^t + \mathbf{C}^{t+1} - \mathbf{V}^{t+1}. \tag{17}$$

Finally, the penalty parameter $\mu$ is updated as follows:

$$\mu^{t+1} = \min(\rho\mu^t, \mu_{\max}). \tag{18}$$

The above three steps are repeated until convergence conditions are reached. The convergence conditions are defined as

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**Algorithm 2** Optimization Procedure for the ETVEBD Algorithm

**Input:**

$\mathbf{X}$: Two dimensional hyperspectral image; $\mathbf{D}$: endmember background dictionary; $\lambda$: the regularization parameter; $\beta$: the regularization parameter.

**Output:**

Anomaly detection map obtianed by (19).

**Initialize:**

all variables $\mathbf{C}$, $\mathbf{V}$ and $\mathbf{M}$ are set to zero matrices, $\mu_0 = 1\mathrm{e}-6$, $\mu_{\max} = 1\mathrm{e}6$, $\rho = 1.1$, $\varepsilon = 1\mathrm{e}-6$, $t = 0$.

1: **while** not converged **do**
2: Update $\mathbf{C}$ and $\mathbf{S}$ with (12) and (14), respectively.
3: Update $\mathbf{V}$ with (16).
4: Update $\mathbf{M}$ with (17).
5: check the convergence condition $\|\mathbf{C}-\mathbf{V}\|_F < \varepsilon$ or $t > T$
6: $t \leftarrow t+1$.
7: **end while**

$\|\mathbf{C}^t - \mathbf{V}^t\|_F^2 \le \varepsilon$ or the max iteration times $T = 400$. Once the optimization for problem (1) is finished, the detection value of each PUT can be calculated based on the minimizer $\mathbf{S}^*$ as follows:

$$\|[\mathbf{S}^*]_{:,i}\|_2 > \eta \tag{19}$$

where $\eta$ is a predefined threshold. The proposed method can be summarized as Algorithm 2.

## III. Experiment Results

In this section, four hyperspectral datasets are utilized to evaluate the effectiveness of the ETVEBD method. All the experiments were conducted on the computer with an Intel i7-11700k CPU and 32-GB RAM. The receiver operating characteristic (ROC) curve and the area under the ROC curve (AUC) value are used to assess the detection performance of the ETVEBD method and compared methods qualitatively and quantitatively [48].

### A. Experimental Datasets

The performance of the ETVEBD is evaluated on four datasets, which are collected at different scenes. The experiment datasets are briefly introduced as follows.

The first dataset (hereinafter referred to D1) and the second dataset (hereinafter referred to D2) were captured in Los Angeles, USA. They were collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor with the spatial resolution of 7.1 m [49]. These two datasets both have a size of $100 \times 100$ pixels and 205 spectral bands. The wavelengths' range of the D1 dataset and the D2 dataset is from 400–2500 nm. The anomaly targets of these two datasets are buildings. The false-color image and ground-truth map of the D1 dataset are shown in Fig. 3(a). The false-color image and ground-truth map of the D2 dataset are shown in Fig. 3(b).

The third dataset (hereinafter referred to as D3) was also captured by the AVIRIS sensor [50]. It has 224 spectral channels over wavelengths ranging from 366 to 2496 nm. The scene of this dataset is EI Segundo, CA, USA, and its spatial resolution is 7.1 m. This dataset is mainly composed of an area of some residential areas, parks, oil refineries, and a school zone. The storage tanks and towers are labeled as anomaly targets, and its false-color image and ground-truth map are shown in Fig. 3(c).

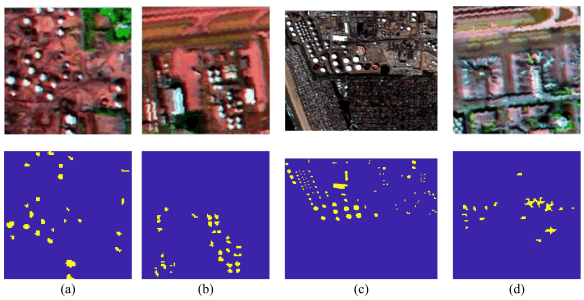


Fig. 3. False-color images and ground-truth maps of four datasets. (a) D1 dataset. (b) D2 dataset. (c) D3 dataset. (d) D4 dataset.

The fourth dataset (hereinafter referred to as D4) was collected at the Los Angeles Airport [49]. It was also captured by the AVIRIS sensor, and its spatial resolution is 7.1 m. It also has 205 bands ranging from 400 to 2500 nm, and the anomaly targets of this dataset are airplanes. Its false-color image and ground-truth map are shown in Fig. 3(d).

### B. Compared Methods and Parameter Settings

To comprehensively test the performance of the ETVEBD method, both classical and state-of-art methods were employed as the compared methods. They are RX [10], CRD [17], RPCA-RX [22], LSMAD [24], LRASR [26], GTVLRR [31], and ADLR [27], respectively.

The RX method, as the benchmark hyperspectral anomaly detection method, is based on the multivariate Gaussian model of background. The CRD is a classical method that is based on the representation model. It has three parameters: the inner window size $w_{\text{in}}$, the outer window size $w_{\text{out}}$, and the regularization parameter $\lambda$. The RPCA-RX detector models the backgrounds by the low-rank constraint and anomalies by the sparse constraint, and applies the RX detector on the sparse component to detect anomalies. The LSMAD detector explores the low-rank prior knowledge of the background to compute the background statistics and applies the Mahalanobis distance differences to detect the anomalies. It has two parameters: the rank $r$ and the cardinality $k$. For convenience, the rank $r$ is also estimated by the Hysime algorithm in the experiments. The LRASR method employs a dictionary strategy and low-rank prior to model the background and uses $l_2$-norm of the sparse component as the detection value. It has four parameters: the cluster number $K$, which is automatically estimated by the Hysime algorithm for convenience, the number of pixels is selected in each cluster $P$, and the tradeoff parameters: $\lambda$ and $\beta$. The GTVLRR method incorporates graph regularization and TV regularization into the LRR model to preserve the local geometrical structure and spatial relationships of the HSI. It has five parameters: the cluster number $K$, which is also automatically estimated by the Hysime algorithm for convenience, the number of pixels that are selected in each cluster $P$, and the tradeoff parameters: $\lambda$, $\beta$, and $\gamma$. The ADLR

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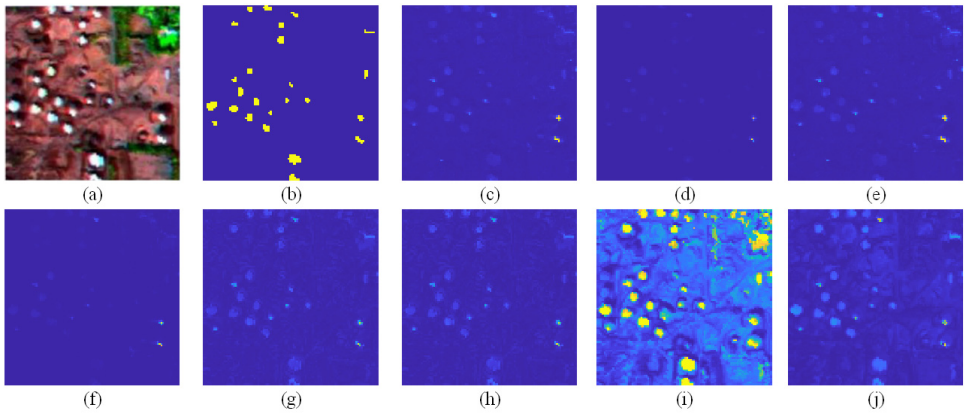


Fig. 4. False-color image, ground-truth map, and 2-D detection maps of compared methods on the D1 dataset. (a) False-color image. (b) Ground-truth map. (c) RX method. (d) CRD method. (e) RPCA-RX method. (f) LSMAD method. (g) LRASR method. (h) GTVLRR method. (i) ADLR method. (j) ETVEBD method.

TABLE I
OPTIMAL PARAMETER SETTINGS FOR THE TESTED METHODS

| Methods | Parameters | Data Sets | | | |
|---|---|---|---|---|---|
| | | D1 | D2 | D3 | D4 |
| CRD | $w_{in}$ | 11 | 5 | 15 | 15 |
| | $w_{out}$ | 13 | 9 | 31 | 17 |
| | $\lambda$ | $10^{-6}$ | $10^{-6}$ | $10^{-6}$ | $10^{-6}$ |
| LSMAD | $r$ | 34 | 25 | 36 | 29 |
| | $k$ | 0.4 | 0.6 | 0.1 | 1 |
| LRASR | $\lambda$ | $10^{-1}$ | $10^{-2}$ | 1 | $10^{-2}$ |
| | $\beta$ | $10^{3}$ | $10^{3}$ | $10^{3}$ | $10^{-1}$ |
| | $K$ | 34 | 25 | 36 | 29 |
| | $P$ | 20 | 20 | 20 | 20 |
| GTVLRR | $\lambda$ | $10^{-3}$ | $10^{2}$ | $10^{-1}$ | $10^{-3}$ |
| | $\beta$ | $10^{2}$ | $10^{-3}$ | $10^{-3}$ | 10 |
| | $\gamma$ | $10^{-1}$ | $10^{-1}$ | $10^{-1}$ | $10^{-1}$ |
| | $K$ | 34 | 25 | 36 | 29 |
| | $P$ | 20 | 20 | 20 | 20 |
| ADLR | $\lambda$ | $10^{-2}$ | $10^{-2}$ | $10^{-1}$ | $10^{-1}$ |
| | $bw$ | 0.9 | 0.3 | 0.7 | 0.5 |
| ETVEBD | $\lambda$ | $10^{-2}$ | $10^{-10}$ | $10^{-4}$ | $10^{-10}$ |
| | $\beta$ | $10^{-10}$ | $10^{-10}$ | $10^{-4}$ | $10^{-10}$ |
| | $p$ | 0.1 | 0.3 | 0.1 | 0.5 |

method is also a method based on the LRR model, while it applies spectral unmixing to obtain the abundance vectors and uses these vectors to build the LRR model. It has only two parameters: the regularization parameter $\lambda$ and the bandwidth parameter bw. The optimal parameter settings for the tested methods are listed in Table I.

### C. Evaluation Criteria

The most widely used evaluation criteria for hyperspectral anomaly detection are the ROC curve and the AUC value. To introduce the ROC curve clearly, the probability of detection (PD) and FAR should be defined. Given a detection map, the ground-truth map, and a fixed threshold value, the PD and FAR can be calculated as follows:

$$\mathrm{PD} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{20}$$

$$\mathrm{FAR} = \frac{\mathrm{FP}}{\mathrm{FP}+\mathrm{TN}} \tag{21}$$

where TP is the anomalies that are truly predicted as anomalies, FN is the number of anomalies that are wrongly predicted as backgrounds, TN is the number of backgrounds that are correctly predicted, and FP is the number of backgrounds that are falsely predicted as anomalies.

With PD and FAR in hand, the ROC can be plotted under different threshold values. If a method has a higher PD than the rest methods with the same FAR, it means that the method has a better performance than others. However, the ROC curve cannot quantitatively evaluate the performance of different methods. To this end, the AUC value, which is the area under the ROC curve, is used to judge the performance of different methods. The method with the highest AUC value has the best performance among different methods. Given a ground-truth map and a detection map, the AUC value can be calculated as follows:

$$\mathrm{AUC} = \int_{-\infty}^{+\infty} \mathrm{PD}(H)\mathrm{FAR}'(H)dH \tag{22}$$

where $\mathrm{PD}(H)$ is the PD when the threshold for the detection map is set as $H$ and $\mathrm{FAR}(H)$ is the false positive rate when the threshold for the detection map is set as $H$.

### D. Experiment Results and Analysis

To evaluate the performance of the ETVEBD method comprehensively, a mass of experimental results of the ETVEBD method and compared methods are discussed in this part. Figs. 4–7 illustrate the 2-D detection maps of different methods on all datasets. As shown in Fig. 4, all the methods except the CRD detector and the LSMAD detector can detect the most anomalies well. The ETVEBD method and ADLR can outline the boundaries clearly, while other methods cannot. In terms of Figs. 5–7, it can also be obviously seen that

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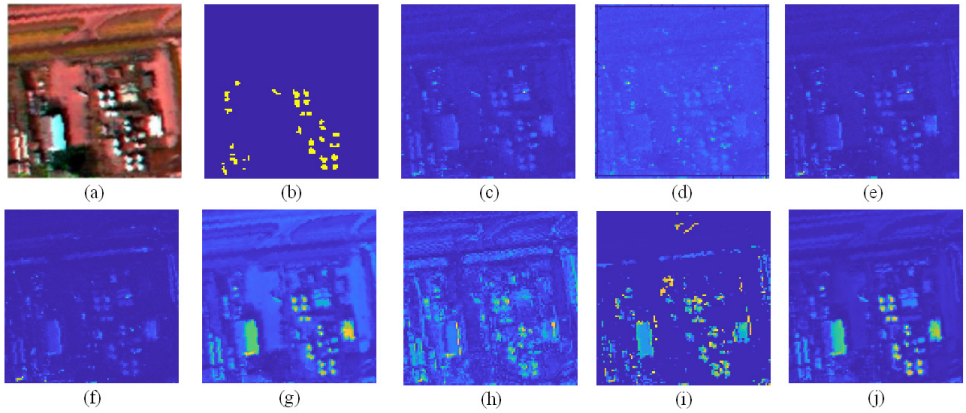


Fig. 5. False-color image, ground-truth map, and 2-D detection maps of compared methods on the D2 dataset. (a) False-color image. (b) Ground-truth map. (c) RX method. (d) CRD method. (e) RPCA-RX method. (f) LSMAD method. (g) LRASR method. (h) GTVLRR method. (i) ADLR method. (j) ETVEBD method.

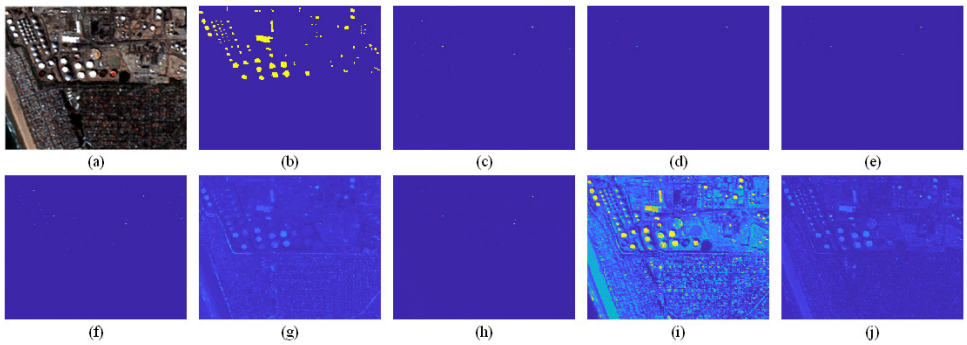


Fig. 6. False-color image, ground-truth map, and 2-D detection maps of compared methods on the D3 dataset. (a) False-color image. (b) Ground-truth map. (c) RX method. (d) CRD method. (e) RPCA-RX method. (f) LSMAD method. (g) LRASR method. (h) GTVLRR method. (i) ADLR method. (j) ETVEBD method.

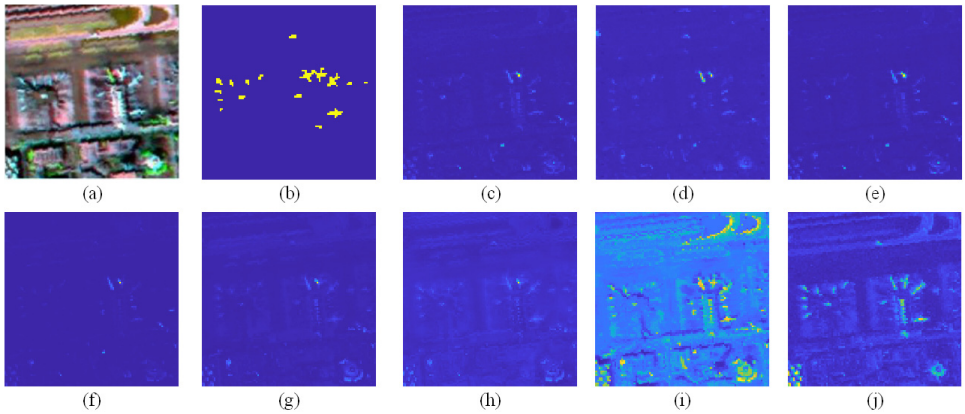


Fig. 7. False-color image, ground-truth map, and 2-D detection maps of compared methods on the D4 dataset. (a) False-color image. (b) Ground-truth map. (c) RX method. (d) CRD method. (e) RPCA-RX method. (f) LSMAD method. (g) LRASR method. (h) GTVLRR method. (i) ADLR method. (j) ETVEBD method.

the ETVEBD method has a better performance than other methods. In summary, the ETVEBD method can detect more anomalies compared to other methods except for the ADLR by the visual analysis. Although the ADLR method can highlight the anomalous targets very well, it cannot suppress the backgrounds very well, which means that it has a higher FAR. However, the ETVEBD method suppresses the background pixels while detecting a lot of anomaly pixels. In order to quantitatively evaluate the performance of the ETVEBD method, ROC curves and AUC values of all methods on four datasets are plotted and listed, respectively. As shown in Fig. 8, it can be seen that the ROC curve of the proposed ETVEBD method is always above those of other anomaly detection methods on four datasets. Table II shows the AUC

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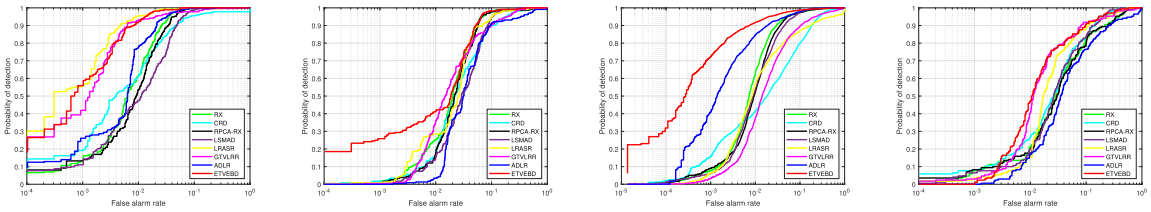
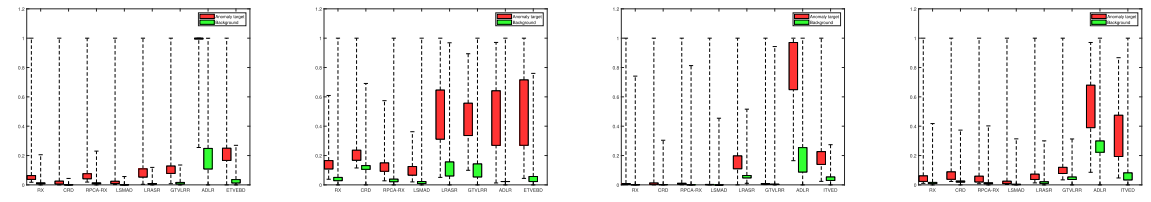


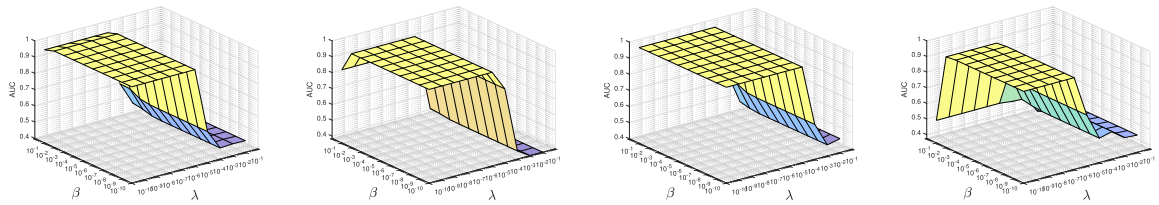

(a) (b) (c) (d)

Fig. 8. ROC curves of the ETVEBD method and compared methods on the datasets. (a) D1 dataset. (b) D2 dataset. (c) D3 dataset. (d) D4 dataset.



(a) (b) (c) (d)

Fig. 9. Separability maps of the ETVEBD method and compared methods on the datasets. (a) D1 dataset. (b) D2 dataset. (c) D3 dataset. (d) D4 dataset.




(a) (b) (c) (d)

Fig. 10. AUC value of the ETVEBD method with the change of $\lambda$ and $\beta$ on the four datasets. (a) D1 dataset. (b) D2 dataset. (c) D3 dataset. (d) D4 dataset.

TABLE II
AUC VALUES OF THE ETVEBD METHOD AND OTHER COMPARED METHODS ON FOUR DATASETS

| Methods | Data Sets | | | |
|---|---|---|---|---|
| | D1 | D2 | D3 | D4 |
| RX | 0.9889 | 0.9694 | 0.9835 | 0.9264 |
| CRD | 0.9680 | 0.9523 | 0.9357 | 0.9495 |
| RPCA+RX | 0.9865 | 0.9664 | 0.9810 | 0.9238 |
| LSMAD | 0.9814 | 0.9583 | 0.9804 | 0.9480 |
| LRASR | 0.9850 | 0.9644 | 0.9259 | 0.9515 |
| GTVLRR | 0.9919 | 0.9610 | 0.9494 | 0.9555 |
| ADLR | 0.9905 | 0.9325 | 0.9905 | 0.8930 |
| ETVEBD | **0.9941** | **0.9755** | **0.9944** | **0.9625** |

values of the ETVEBD method and compared methods on the four datasets, and the highest AUC value on each dataset is highlighted in bold. It can be clearly found that the ETVEBD method has the highest AUC value on each dataset, and the AUC values of the ETVEBD method are 0.9941, 0.9755, 0.9944, and 0.9625, respectively.

To further elaborate the separability performance of the ETVEBD method, the box graphs of the ETVEBD method and other compared methods on four datasets are displayed in Fig. 9. As shown in Fig. 9, there are two columns plotted for each method, i.e., the anomaly column and the background column. In addition, the red and green boxes represent the distribution of the detection values of the anomalies and the backgrounds, respectively. In every column, the detection values of all methods are normalized to the range of 0–1. Thus, the positions of these boxes demonstrate the separability between anomalies and backgrounds. Moreover, the upper and lower edges of boxes depict the tenth and 90th percentiles of the detection values, respectively. The black line in each box indicates the median value of the detection value for each method on different datasets. The whiskers of every box present the max value and the min value, respectively. For the D1 dataset, the D3 dataset, and the D4 dataset, as shown in Fig. 9(a), (c), and (d), the gaps between anomalies and backgrounds for the ETVEBD method and the ADLR method are obviously bigger than the others. As for the D2 dataset, the gaps between anomalies and backgrounds for the ETVEBD method are nearly the same as the ADLR whose gaps between anomalies and backgrounds are the biggest. Therefore, it can be concluded that the separability between anomalies and backgrounds of the ETVEBD method is excellent.

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TABLE III
RUNNING TIMES (SECONDS) OF THE ETVEBD METHOD AND OTHER COMPARED METHODS ON FOUR DATASETS

| Methods | Data Sets | | | |
|---|---|---|---|---|
| | D1 | D2 | D3 | D4 |
| RX | **0.5623** | **0.8016** | **3.0506** | **0.7776** |
| CRD | 3.3234 | 6.3926 | 4247.4432 | 5.1521 |
| RPCA+RX | 2.6595 | 3.2379 | 26.3917 | 2.8049 |
| LSMAD | 6.0342 | 125.6067 | 5633.0884 | 126.3943 |
| LRASR | 44.4897 | 39.4225 | 472.3436 | 37.3272 |
| GTVLRR | 73.5475 | 82.0384 | 875.1777 | 71.1346 |
| ADLR | 20.1209 | 268.5739 | 263.8747 | 86.8973 |
| ETVEBD | 23.9358 | 22.5570 | 335.2720 | 27.5266 |

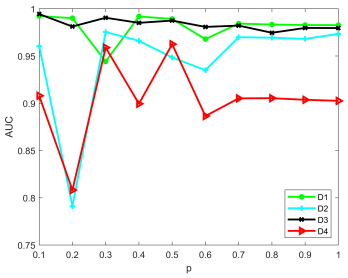


Fig. 11. AUC value of the ETVEBD method with the change of $p$ on the four datasets.

### *E. Computational Efficiency Analysis and Parameters' Sensitivity Analysis*

The running time of the ETVEBD method and compared methods on the four datasets can be found in Table III. As shown in this table, because the RX detector just needs to calculate the background variance matrix and the background mean vector, its running time is the least. The running time of the CRD detector is closely related to the outer window size $w_{\text{out}}$ and the inner window size $w_{\text{in}}$. Due to the fact that the difference between the outer window size $w_{\text{out}}$ and the inner window size $w_{\text{in}}$ is extremely large, the CRD detector on the D3 dataset also takes a long time to run. Although the running time of the ETVEBD is not the least, it is much less than the algorithms, such as LRASR, GTVLRR, and ADLR, which are all the representation model methods. The reason for this phenomenon may be that the EBD's atom number is usually very small in the ETVEBD method. It also indicates that the proposed EBD has better representation capability than the traditional background dictionary.

The parameter sensitivity of the ETVEBD method is also analyzed in this part. The ETVEBD method totally has three parameters, that is, the tradeoff parameters $\lambda$ and $\beta$ and the percentiles of the endmembers selected to form the endmember dictionary $p$. Fig. 10 reveals the influence of different trade parameters $\lambda$ and $\beta$ on the AUC value on the four datasets. It is clear that the influence of the trade parameters $\lambda$ to the AUC value on the four datasets is greater than $\beta$. However, the AUC value varies very little over a large range of values for $\lambda$.

TABLE IV
AUC VALUES OF THE ETVEBD METHOD AND THE TVEBD METHOD ON FOUR DATASETS

| Methods | Data Sets | | | |
|---|---|---|---|---|
| | D1 | D2 | D3 | D4 |
| ETVEBD | **0.9941** | **0.9755** | **0.9944** | **0.9625** |
| TVEBD | 0.9932 | 0.9511 | 0.9941 | 0.8808 |

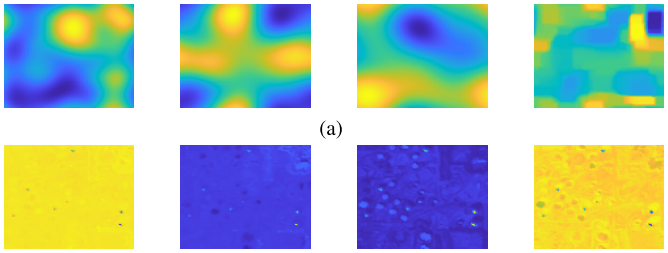





Fig. 12. Representation coefficient images of the D1 dataset based on the TV term regularized representation model and the ETV term regularized representation model. (a) Representation coefficient images of the TV term regularized representation model. (b) Representation coefficient images of the ETV term regularized representation model.

Therefore, a conclusion may be drawn that the sensitivity of the ETVEBD to the tradeoff parameters $\lambda$ and $\beta$ is acceptable.

Besides, the AUC value of the ETVEBD method with the change of parameter $p$ is shown in Fig. 11. When the AUC value of the ETVEBD method is the highest, the values of $p$ are 0.1, 0.3, 0.1, and 0.5, respectively. These results show that the DPC method can efficiently avoid the anomaly endmembers to be selected as the dictionary atoms.

### *F. Effectiveness of the Proposed ETV Term*

In addition, to show the effectiveness of the proposed ETV term, the comparative experiment is also conducted on all four datasets. The TV regularized representation model with EBD (TVEBD) that uses the same endmember dictionary as ETVEBD is selected as the compared method. The AUC values of the ETVEBD method and the TVEBD method on four datasets are listed in Table IV. It can be clearly found from Table IV that the AUC value of the ETVEBD is larger than TVEBD. Furthermore, to intuitively demonstrate the effectiveness of the ETV term, the representation coefficient images of the D1 dataset that is selected as the example are plotted in Fig. 12. Concretely, the representation coefficient images are plotted by reshaping the rows of representation coefficient matrix $\mathbf{C}$. As shown in Fig. 12(a), it can be seen that the representation coefficient images based on the TV regularized representation model cannot outline the boundaries of the D1 dataset. Nevertheless, the representation coefficient images based on the ETV term regularized representation model can highlight the boundaries of the D1 dataset, as shown in Fig. 12(b). That is to say, the proposed ETV term can enhance the spatial structure of an HSI in the representation process.

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## IV. Conclusion

In this article, a novel ETV term regularized representation model with the endmember dictionary method for hyperspectral anomaly detection is proposed. Different from the original TV term that is widely used in traditional representation models, the proposed ETV term can enhance the spatial structure of an HSI. Meanwhile, as the existence of mixed pixels due to the fact that the spatial resolution of an HSI is relatively low, anomaly components are inevitably mixed into the background dictionary, which can deteriorate the performance of the representation-based method. Thus, a new EBD construction method based on VCA and DPC algorithms is proposed. The new dictionary construction method can preserve the physical meanings of the representation coefficients while reducing the influence of anomaly components in mixed pixels. To verify the effectiveness of the ETVEBD method, other seven classical and state-of-art methods are chosen as the comparative algorithms. Extensive experiments on four datasets exhibit the superiority of the proposed ETVEBD method.

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